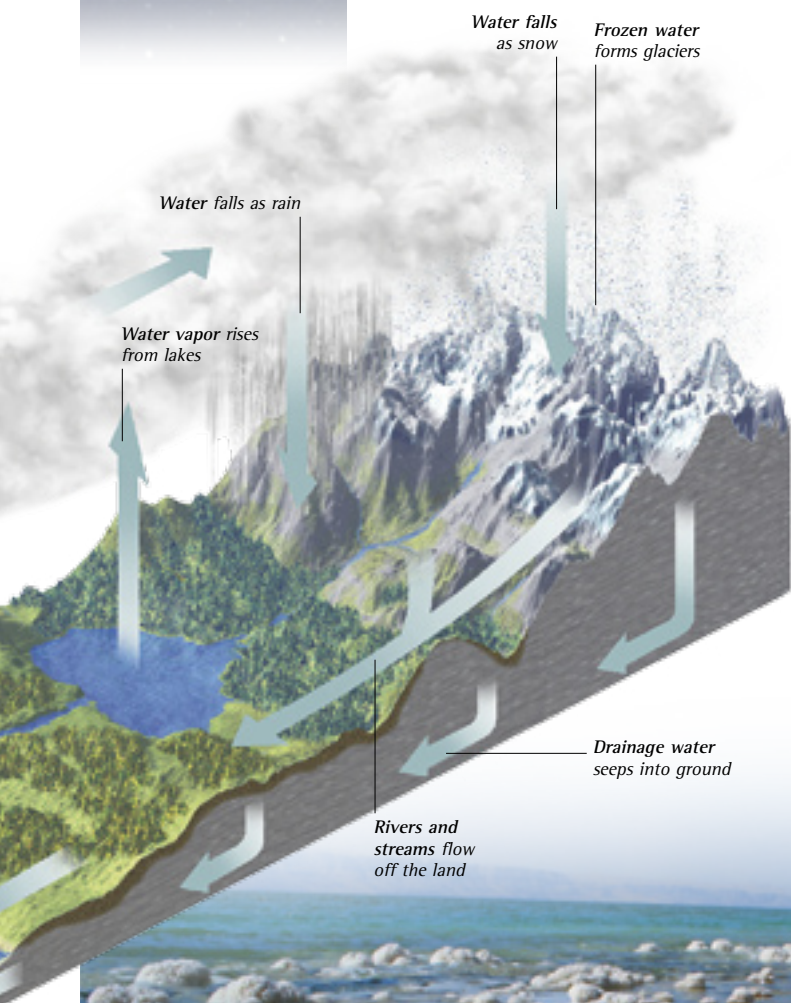


◀ WATER FROM SPACE

The comets that regularly enter the solar system are made of ice and dust. Hence, they are often described as "dirty snowballs." It is likely that many comets struck Earth during its early history, and the water from these comets melted and flowed into the oceans. Some scientists believe that this water from space may have contained complex molecules that formed the seeds of life on Earth.



◀ THE WATER CYCLE

The sun-warmed surface water of the oceans is constantly evaporating, or turning into water vapor that rises into the air. Here, the vapor cools and condenses into clouds, which also contain other gases dissolved from the atmosphere. The clouds release rain that often falls on land. Water then flows off the land back to the sea, carrying dissolved minerals and sand, silt, and mud particles into the ocean.

▼ OCEAN CHEMISTRY

Water is very good at dissolving minerals and gases, which then form a part of its chemistry. These include carbon, oxygen, nitrogen, phosphorus, calcium, and iron—substances that are vital to animals, such as this crayfish, for tissue building and fueling body processes.

Many of these substances remain on the ocean floor until the sediments are stirred up by strong currents.



▼ THE SALTY SEA

The water flowing off the land in rivers contains dissolved minerals called salts. The salt content in river water is so low that we think of it as fresh water, but in hot regions it often evaporates to leave thick salt deposits, like these at the edge of a salt lake. This occurs in the oceans all the time, and over billions of years, it has made ocean water salty. Most of the salt is sodium chloride, which is the same mineral as table salt.

